

2023



GE VERNOVA

Rightsizing and Optimizing Infrastructure to Cut \$1 Million in Costs with GE Vernova

Learn how GE Vernova boosted compute performance with significant cost savings using AWS Graviton processors.

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\$1 million

in overall cost savings achieved in less

\$100,000

saved by decommissioning idle instances



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I \$600,000+



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\$64,000

saved in removal of manual snapshots



80%

nonproduction instances configured for \$460,000 yearly savings

Overview

[GE Vernova](#) was looking to optimize and cut significant costs from its cloud operations as it prepared to become an independent company. Specifically, GE Vernova wanted to streamline its use of [Amazon Relational Database Service](#) (Amazon RDS), a collection of managed services that makes it simple to set up, operate, and scale databases in the cloud.

GE Vernova worked alongside Amazon Web Services (AWS) to identify potential areas of savings within [Amazon Elastic Compute Cloud](#) (Amazon EC2), which provides secure and resizable compute capacity for virtually any workload. Using AWS tools for greater visibility, engineers cut more than \$1 million by incorporating automation and other best practices in life cycle management, database optimization, and rightsizing of infrastructure.



Opportunity | Using AWS Tools to Drive Cost Savings for GE Vernova

GE Vernova is accelerating the path to more reliable, affordable, and sustainable energy while  partners to power economies and deliver the electricity that is vital to health, safety, security, and improved quality of life.

GE Vernova's technology helps to generate approximately 30 percent of the world's electricity, supporting customers through the energy transition. Its parent company, General Electric, had been using AWS since 2012. As GE Vernova prepared to become an independent company sometime in early 2024, the database team looked at areas for cost savings and improvements. "We implemented reporting for all our individual accounts, but we needed to pull it all together and continually refactor it so that it was accurate," says Neil Leslie, database operations leader for GE Vernova. "It's all about having good vision and understanding the data so that you can act on it."

To monitor individual accounts, GE Vernova uses [AWS Trusted Advisor](#), which provides recommendations that help organizations to follow AWS best practices. The database team gains visibility into these findings through [Amazon CloudWatch](#), which collects and visualizes near-real-time logs, metrics, and event data in automated dashboards to streamline infrastructure and application maintenance. Working closely alongside AWS technical account managers, specialists, and operational support teams, GE Vernova defined specific work streams across multiple accounts where it could save significant costs. "At the start of the project, AWS provided a huge amount of support for getting the data required and recommending best practices for resizing, which stopped us from making mistakes," says Leslie. "I haven't seen that all-encompassing service from a vendor for a long time. I know AWS is customer-oriented, but still, this was very impressive and one of the reasons for the project's success."

Solution | Converting to AWS Graviton Processors for over \$600,000 in Savings at Top Price Performance

Initially, GE Vernova focused on low-risk areas for immediate cost savings. The team cut \$64,000 by removing manual snapshots and automating the process through [AWS Lambda](#), a serverless, event-driven compute service. Next, the team automated the shutdown of about 80 percent of its nonproduction Amazon RDS instances when not in use, configuring them using [AWS Instance Scheduler](#), which configures start and stop schedules for Amazon EC2 and Amazon RDS instances to manage costs. Using AWS Instance Scheduler, GE Vernova saves about \$460,000 yearly.

Using data from AWS Trusted Advisor and Amazon CloudWatch, the team identified instances that had been idle for long periods, and it automated communications to the associated application teams to confirm that it was OK to decommission them. The initial implementation saved about \$100,000 and continues to generate small monthly savings.



In late 2022, GE Vernova focused on resizing initiatives, using data from [Amazon RDS Performance Insights](#), which analyzes and tunes Amazon RDS database performance, along with Amazon CloudWatch and internal database tools. Seeking to save on costs and modernize its compute infrastructure, the team saved \$504,000 by downsizing 50 instances to smaller instance types. It also migrated 128 instances to the second generation of [AWS Graviton](#) processors, a family of processors designed to deliver the best price performance for cloud workloads running on Amazon EC2. “With the number of instances we’ve migrated, it’s been quite striking that we haven’t had significant performance issues,” says Leslie. “We see a very reliable processor that helps us downsize capacity while gaining a performance boost.” In 2023, the team converted an additional 246 instances to AWS Graviton2 processors for another \$100,000 worth of savings in 3 months. The team projects additional savings of \$200,000 per year by implementing [Amazon EC2 Reserved Instances](#), which provide a discount of up to 72 percent compared with Amazon EC2 On-Demand Pricing and provide a capacity reservation when used in a specific Availability Zone.

In addition to cost-saving measures in its database applications, GE Vernova saves \$18,000 per month by optimizing storage through its use of [Amazon Simple Storage Service](#) (Amazon S3), an object storage service offering cutting-edge scalability, data availability, security, and performance. The company configured [Amazon S3 Lifecycle](#) policies to define when to archive objects by migrating them to [Amazon S3 Glacier storage classes](#), which provide long-term, secure, durable storage classes for data archiving at the lowest cost and milliseconds access.

Outcome | Saving \$1 Million While Optimizing Infrastructure Companywide

In less than a year, GE Vernova met its million-dollar target. By eveloping automations and applying AWS best practices to its infrastructure, the team modernized its traditional approach to database management and is applying new skills to future solutions. Additionally, the team is sharing its learnings across all GE Vernova so that other teams can benefit from best practices in cost cutting and modernization.

“Using AWS helps us put the entire business on a better path,” says Leslie. “We optimize and deliver cutting-edge solutions that help GE Vernova meet its business goals to continue to electrify and decarbonize the world. That’s the end game.”

About GE Vernova

GE Vernova is a planned, purpose-built global energy company that includes power, wind, and electrification segments and is supported by its accelerator businesses of advanced research, consulting services, and financial services. It serves customers in 140 countries.

AWS Services Used

Amazon RDS



Amazon Relational Database Service (Amazon RDS) is a collection of managed services that makes it simple to set up, operate, and scale databases in the cloud.

[Learn more »](#)

Amazon EC2

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 750 instances and choice of the latest processor, storage, networking, operating system, and purchase model to help you best match the needs of your workload.

[Learn more »](#)

AWS Graviton processor

AWS Graviton is a family of processors designed to deliver the best price performance for your cloud workloads running in Amazon Elastic Compute Cloud (Amazon EC2). Choose the AWS Graviton-based instance that best meets your needs.

[Learn more »](#)

Amazon CloudWatch

Amazon CloudWatch is a service that monitors applications, responds to performance changes, optimizes resource use, and provides insights into operational health.

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